

WATSON AGBRAMU

Senior Software Engineer

watsonagbramu@gmail.com

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Ontario, Canada

SUMMARY

Software Engineer with 8+ years building production systems across SaaS, media, and fintech, from client-facing APIs and interfaces through data pipelines and cloud infrastructure. Recent work in LLM-powered products, multi-tenant platforms, and event-driven architecture.

SKILLS

Languages	Python, TypeScript, JavaScript, SQL, Rust
AI	LangChain, LangGraph, LiteLLM, OpenAI SDK, Anthropic SDK, RAG, vector search, embeddings, agentic workflows, tool calling, evals
Backend	FastAPI, Flask, Node.js, Express, GraphQL, REST, Celery, event-driven architecture
Frontend	React, Next.js, React Native, Tailwind CSS
Data	PostgreSQL, MongoDB, MySQL, Redis, Vector DB, DBT, Snowflake, Airbyte, Databricks
Infra	AWS (Lambda, SQS, S3, CloudFront, Route 53, SAM), GCP, Azure, Docker, Terraform, ArgoCD, Argo Workflows, GitHub Actions

EXPERIENCE

Senior Software Engineer

Sept 2024 – Present

LipDub AI (MARZ)

Remote

- Initiated and designed the personalization epic, delivering the API in production today that regenerates voice and lip-sync for edited text segments, eliminating full-asset re-renders on client revisions
- Built and own the public video-generation API and the internal service layer behind the Next.js frontend using Python, Flask, and FastAPI, covering generation, asset management, and account workflows
- Architected feature gating on Stripe entitlements as the plan-level source of truth with LaunchDarkly as a per-tenant resolver, avoiding a bespoke entitlements service, with short-TTL caching invalidated on Stripe plan-change webhooks to keep gate checks off the critical path
- Shipped custom subdomain support on AWS CloudFront and Route 53, automating CNAME delegation and TLS provisioning while scoping edge routing to expose only client-facing watch routes
- Built a media ingestion worker accepting public and YouTube URLs and added workflow prioritization for Argo generation jobs, authoring the workflow configuration and moving download and validation server-side
- Delivered voice cloning, occlusion handling, AI avatars, Video looping, translation memory, and the assets library across Python services and Next.js, deployed via ArgoCD

Senior Software Developer

Feb 2022 – Sept 2024

Arctic AI

Remote

- Developed RAG-based AI systems using LangChain and LLMs in production workflows, tuning chunking and retrieval strategy against domain-specific query patterns
- Implemented vector search infrastructure for semantic retrieval, improving answer grounding and reducing token spend by narrowing context to relevant messages
- Built an enterprise data platform on Azure (Databricks, Data Factory, Data Lake), consolidating overlapping pipelines and scaling multi-source ingestion across client products
- Led a full platform redesign in Next.js and implemented CI/CD with GitHub Actions and Docker, standardizing deployments across services
- Mentored engineers and contributed to system design and technical direction across projects

Web Application Developer

Divergence Neuro

Aug 2021 – Oct 2021

Remote

- Developed a client-facing web application in React and AWS, integrating backend API endpoints and implementing PWA compliance with service workers for offline capability
- Built a custom QR scanner and Bluetooth integration in React, enabling users to connect directly to Neurosity devices

Software Developer

Neo Financial

Jun 2021 – Aug 2021

Remote

- Built credit statement generation for printing at Symcor using Node, React, AWS, Terraform, TypeScript, GraphQL, and MongoDB
- Engineered the decider microservice routing declined transactions to a third-party fraud detection API, and optimized the bank microservice behind MasterCard payment gateway integration

Data Scientist (M.Eng Research)

University of Waterloo

Sept 2019 – Dec 2020

Waterloo, ON

- Built image classification models using transfer learning (Inception-ResNet-V2, EfficientNet), reaching 92.18% accuracy across 120 classes
- Improved pedestrian detection by re-annotating datasets and training Faster R-CNN models with optimized data pipelines

PROJECTS

Natural-Language Business Intelligence Platform

FastAPI, LangGraph, Next.js

- Built a multi-tenant BI platform where a LangGraph agent translates plain-English questions into SQL across Postgres, MySQL, SQLite, and Snowflake, selects a visualization, and assembles shareable dashboards
- Added a semantic business-term layer, natural-language alerts, automated anomaly and root-cause analysis, and usage-based Stripe billing, with Celery and Redis running scheduled and long-running jobs

On-Device Voice Dictation (macOS, Android)

Rust, Swift, Kotlin

- Built push-to-talk dictation that transcribes speech to the cursor in any application, running recognition and cleanup entirely on device so no audio leaves the machine at runtime
- Implemented one Rust core (whisper.cpp, Silero VAD, session state machine) exposed through UniFFI to native shells: a Swift menu-bar app on macOS and a Kotlin input method on Android

AI-Assisted Strength Training App

React Native, TypeScript, FastAPI

- Built an offline-first mobile app (React Native, Expo, SQLite, Supabase, FastAPI) converting a conversational intake into a multi-week training program that auto-progresses weights from logged set performance
- Constrained LLM calls to the edges of a deterministic training engine, validating every model output against program invariants before it can modify a plan

EDUCATION

M.Eng, Electrical and Computer Engineering (Distinction), University of Waterloo Sept 2019 – Dec 2020

B.Eng, Electrical and Electronics Engineering (First Class), Federal University Oye Feb 2013 – Jan 2018